Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/14/2017
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Project Title:	Determination of Roles of Microgravity and Ionizing Radiation on the Reactivation of Epstein-Barr Virus		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	77058-2720	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	10/01/2013	**End Date:**	10/01/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2018 (using 10/01/2018 as end date) per F. Hernandez/ARC (Ed., 9/20/17) NOTE: Extended to 9/30/2017 per F. Hernandez/ARC (Ed., 9/28/16)		
Key Personnel Changes/Previous PI:	July 2017: no changes		
COI Name (Institution):	Bloom, David Ph.D. (University of Florida) Pierson, Duane Ph.D. (NASA Johnson Space Center) Stowe, Raymond Ph.D. (Microgen, LLC) Wu, Honglu Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We will study the effects of simulated microgravity and ionizing radiation on the complex process of virus reactivation that is intimately associated with the way cells respond to external stress. Tools to evaluate the effects of radiation and simulated microgravity will include: both viral and cellular gene expression, viral infectivity and viral titer, and DNA methylation. In addition, this proposal aims to increase our understanding of mechanisms of Epstein-Barr Virus (EBV) reactivation and therefore, has the potential to discover new therapeutic targets for diseases caused in the general population by EBV. Therefore this proposal appropriately addresses: 1. Ground-based Research Designed to Lead to Space Flight; 2. Cell Biology and Microbiology: This section of the NASA Research Announcement (NRA) solicits NOVEL ground-based experiments that use cell cultures, tissue cultures, microbiological organisms, or small invertebrate organisms to elucidate the basic mechanisms living systems use to respond to changes in gravity, i.e., how are changes sensed and transduced to alter gene expression and the structure and function of the musculoskeletal system. The reactivation of all herpes viruses is known to involve 2 components or stages: the initial initiating molecular events that cause reactivation of the latent genomes to progress to the production of reactivating virus, and then the propagation of these reactivating viruses to produce clinical disease. The second part of this process undoubtedly involves overcoming components of the acquired immune response if clinical disease is to result. In order to understand reactivation from a mechanistic point of view, these two processes need to be examined independently. In the proposed studies, we will be focusing only on the initial events that microgravity and ionizing radiation initiate in latent cells to induce the first stage of (molecular) reactivation. EBV reactivation may not be completely homologous to Varicella zoster virus (VZV) and Cytomegalovirus (CMV) reactivation. The important point here though is that by focusing on EBV, which has a well-established cell culture system of latency, we can study the mechanistic effects of microgravity and ionizing radiation in inducing EBV reactivation. This alone will provide critical and important information regarding the effects of these inducers on EBV reactivation in humans. Clearly, after these molecular events are established in EBV, possible parallel mechanisms in VZV and CMV can be investigated in future studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to increase our understanding of mechanisms of Epstein-Barr Virus (EBV) reactivation and therefore, has the potential to discover new therapeutic targets for diseases caused in the general population by EBV.
	Aim 1: To optimize the protocols for the experiments at Johnson Space Center (JSC) and Brookhaven National Laboratory (BNL). Aim 2: To culture the Akata EBV Positive cells in microgravity conditions. Aim 3: To culture the Akata EBV Positive cells after being exposed to gamma-radiation at varying doses in T-flasks. Aim 4: To culture the Akata EBV Positive cells after being exposed to gamma-radiation at varying doses and microgravity conditions. Aim 5: To culture the Akata EBV positive cells after being exposed to proton radiation at varying doses in T-flasks. Aim 6: To culture the Akata EBV positive cells after being exposed to proton radiation at varying doses and microgravity conditions. Aim 7: To culture the Akata EBV positive cells after being exposed to carbon ion radiation at varying doses in T-flasks. Aim 8: To culture the Akata EBV positive cells after being exposed to carbon ion radiation at varying doses and microgravity conditions. Aim 9: To culture the Akata EBV positive cells after being exposed to iron ion radiation at varying doses in T-flasks. Aim 10: To culture the Akata EBV positive cells after being exposed to iron ion radiation at varying doses and microgravity conditions. Accomplishments towards Aim 1 Cell Culture of Cells in T-Flasks: The Akata EBV Positive cells were aseptically cultured and gradually expanded in T-75 flasks and eventually T-175 flasks using media and supplements per recommendation of Jeffrey Cohen. Cells were incubated at 370C, 5% CO2, and the flasks were kept in an upright position. Cells were split and fed every four days, once the media had turned yellow, by removing 25 ml of cells and media and replacing it with 25 ml of fresh RPMI 1640 complete media. During early passage numbers of the cells, 1.0e7 cell aliquots were frozen down in case anything catastrophic happened to the cell line. Optimizing Viewing Cells and Images: T-75 flasks and bioreactors were seeded with 1.0e6 cells/ml, 50 ml of media, while optimizing the experiment. A small aliquot of cells from each bioreactor were added to a corresponding well on a 24 Well Cell Culture Cluster (Costar, Corning, NY) so that the cells could be viewed and photographed using the Nikon TMS Inverted Microscope for each time point at 40X. Cells in T-75 flasks were able to be viewed and photographed by laying the T-75 flasks horizontally on the microscope. All images taken at 40X were saved and captioned. Optimizing Cell Counts and Viability: For each flask and bioreactor an aliquot of cells were stained at a 1:1 ratio with acridine orange and propidium iodine (AO/PI) and then added to a disposable cell counting chamber (Nexcelom Bioscience, Lawrence, MA). Cell counts, viability, and cell size were evaluated using the Nexcelom designed EBV Assay using a Cellometer Auto 2000. Aliquots of Cells for Other Labs: It had been predetermined that 1.0e6 cells would be used for testing the viral load of EBV at JSC and that 3.0e6 cells would each be sent to David Bloom's lab and Raymond Stowe's lab. From the Cellometer Auto 2000 results, the volumes needed for 1.0e6 cells and 3.0e6 cells were calculated. 3.0E6 cells were aliquoted for both Dr. Raymond Stowe and Dr. David Bloom's labs. The cells were centrifuged for 15 minutes at 40C and 2000 rpm to obtain a pellet. The cell pellet for Bloom's lab was resuspended in 300 ul of RNALater RNA Stabilization Reagent at 40C overnight and were then moved and stored in a -800C freezer (Qiagen, Germantown, MD). The cell pellet for Stowe's lab was resuspended in 200-300 ul of supernatant and was then put in the Mr. Frosty freezing container in the -800C overnight. The next morning the frozen cell suspension was moved out of the freezing container and into a labeled box for Dr. Stowe in the -800C. The 1.0e6 aliquot was for DNA extraction at the JSC lab.

After the necessary aliquots were made, additional aliquots were made to be stored in the -80c so that DNA extraction could be repeated if necessary.

Irradiation of Cells: For irradiation the cells were exposed to one of four different types of radiation: 137Cs gamma, proton, iron ion, or carbon ion on day 0. The cells were exposed to an acute dose of 0.1, 0.5, 1.0, or 2.0 Gy at room temperature. Proton, iron ion, and carbon ion irradiation was performed at the Brookhaven National Laboratory's facilities. 137Cs gamma irradiation was performed at both NASA Johnson Space Center's and Brookhaven National Laboratory's facilities. The energies of gamma, proton, iron ion, and carbon ion beams were 150, 150, 600, and 600 MeV/u, respectively.

Shipment of Cells to Brookhaven National Laboratory: The Akata EBV positive cells underwent several trial runs to optimize the cell viability during shipment to and from Brookhaven National Laboratory (BNL). It was determined that the following methods yielded the best viability for the Akata EBV positive cells. On day 0 cell counts and viability were performed and aliquots were made for each lab as previously described. With fresh media, 70 ml of cells were added at a concentration of 1.0e6 cells/ml to a T-25 flask (Costar, Corning, NY). The flask lids were completely tightened and the bottles were sealed with parafilm before being placed back in the incubator at 370C, 5% CO2. On day 1 the cells were removed from the incubator and shipped at room temperature to Brookhaven National Laboratory in Upton, New York. The cells were received at BNL on day 2 and the cells were put in an incubator at 370C, 5% CO2 to recover from the shipping conditions. On day 3 the cells were taken out of the incubator to be irradiated at room temperature. After irradiation, the cells were moved back to an incubator at 370C, 5% CO2 until day 4 when the cells were shipped back to Johnson Space Center at room temperature. On day 5 the cells were received at JSC and were put in an incubator at 370C, 5% CO2 to recover from the shipping conditions. On day 6 the cells were split, aliquoted, fed, and then transferred to the correct cell culture vessel, either a T-75 flask or a 50 ml High Aspect Ratio Vessel (HARV) bioreactor before being returned to the incubator. For graphing purposes, all BNL time points from day 6 to day 22 were adjusted by two days to align with the time points from the JSC portion of the experiments.

DNA Extraction: For DNA Extraction, the ArchivePure DNA Cell/Tissue kits was used and their DNA Purification from 1-2 million cultures cells protocol was modified and optimized (5Prime, Gaithersburg, MD.) The cell pellet and supernatant were separated via centrifugation at 14,000 RPM for 20 seconds. 200 ul of supernatant was added to a separately labeled microcentrifuge tube and 20 ul of residual liquid was used to resuspend the cell pellet in the original microcentrifuge tube. Both the resuspended cell pellet and the supernatant underwent the DNA extraction protocol. The modifications for this protocol include cooling the sample to room temperature (RT) by placing it in ice for 15 minutes rather than 1 minute at the beginning of the protein precipitation steps, centrifuging for 5 minutes at 14,000 RPM for both of the centrifugation steps for DNA precipitation rather than 1 minute, and incubating for 30 minutes at 650C for rapid DNA hydration rather than for 5 minutes.

Nanodrop and QPCR: To determine the concentration of DNA present in each sample, nanodrop was run on each sample the day of the initial quantitative PCR (qPCR) reaction. If for some reason the samples were run again after the samples had been stored at 40C for several weeks then nanodrop was redone on the samples. For determining viral load, qPCR was done using the Fast Reaction plates on the TaqMan 7900HT Fast Real-Time PCR System (Applied Biosystems, Grand Island, NY). A fluorogenic probe, 5'/6-FAM-TGTACACGCACGAGAAATGCGCC-TAMRA/3', was synthesized from IDT DNA Technologies. The primers, 5'CGGAAGCCCTCTCTGGACTTC-3' and 5'CCCTGTTTATCCGATGGAATG-3', from the BALF5 gene were also synthesized from IDT DNA Technologies. The optimized qPCR reaction mix contained 10% of DNA, 1X TaqMan Master Mix, 200 µM each primer, 150 µM probe, and 43.5% H20. The threshold and baseline number for cycles was set to automatic on the program, and each sample was run in duplicate. A standard curve was obtained using BALF5 standards that were serially diluted, and was used to calculate the viral copy number for each reaction well that was run. Normalizing Data from QPCR: To normalize the data, the mean viral copy number from the sample duplicates was calculated. The mean value from each sample was then divided by the concentration determined by Nanodrop for the same sample. After all of the results were expressed as copies of EBV per ng of DNA the values were then converted to represent the percent change from day 0, thus meaning that all values (day 4, day 8, day 12, day 16, and day 20) from a sample were compared to the day 0 value. From the percent changes, the mean from all of the repeats was calculated and then graphed.

Accomplishments towards Aim 2:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 50 ml of media to a 50 ml high-aspect ratio vessel (HARV) bioreactor (Synthecon, Inc. Houston, TX). The bioreactors were attached to a RCCS base rotator in a 370C, 5% CO2 incubator, and the microgravity environment was modeled by rotating the bioreactors at 10 rotations per minute. There were two different vessel conditions for control 1 x g conditions: an upright T-75 flask with a vented cap (Corning Inc. Corning, NY) and a 50 ml HARV bioreactor in a static horizontal position (Synthecon, Inc. Houston, TX). For the control conditions, the cells were seeded at the same concentration and volume of media, and were grown in normal conditions. Cells were aseptically split and harvested every four days by removing 25 ml of cells and media and replacing it with 25 ml of fresh RPMI 1640 complete media until day 16.

Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. The Akata EBV positive cells did not show any significant change in cell viability when comparing the microgravity conditions to the control conditions. Additionally, the Akata EBV positive cells did not show any significant changes in either total cell size (microns) or live cell size (microns) when comparing the microgravity condition to the control conditions. The Akata EBV positive cell pellet showed a statistically significant decrease in the mean percent change of EBV viral load from day 0 on day 4, day 8, day 12, and day 16 in the modeled microgravity condition; whereas, there was no significant change for the control conditions.

Accomplishments towards Aim 3:

Akata EBV positive cells were irradiated with 137Cs gamma irradiation at both the NASA Johnson Space Center's and Brookhaven National Laboratory's facilities. The cells were exposed to an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy. After irradiation, cells were seeded into an upright T-75 flask as previously described in Aim 2. For the control conditions, cells were grown in a T-75 flask as also previously mentioned in Aim 2. Cells were aseptically split and harvested every four days by removing 25 ml of media and replacing it with 25 ml of fresh RPMI 1640 complete media until day 16 or day 20 for the cells irradiated at the BNL facility.

As mentioned in Aim 2, the Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. The Akata EBV positive cells did not show a significant change in cell

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viability when comparing the different acute radiation doses to the control conditions. Additionally, there were no significant changes across the various time points either. Akata EBV positive cells also did not show any significant changes in either total cell size (microns) or live cell size (microns) when comparing the different acute radiation doses to the control conditions, nor were there any significant changes across the various time points. The Akata EBV positive cell pellet showed a statistically significant increase in the mean percent change of EBV viral load from day 0 for the 0.5 Gy, 1.0 Gy, and 2.0 Gy flasks on day 4, when compared to the control conditions. The Akata EBV positive cell pellet also showed a statistically significant increase in the mean percent change of EBV viral load from day 0 for the 2.0 Gy flasks on day 8, when compared to the control conditions.

Accomplishments towards Aim 4:

As described in Aim 3, the Akata EBV positive cells were exposed to an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy with a 137Cs gamma source at both the NASA Johnson Space Center's and Brookhaven National Laboratory's radiation facilities. The Akata EBV positive cells were seeded into a 50 ml bioreactor as previously described in Aim 2. After irradiation, the bioreactors were put back into a 370C, 5% CO2 incubator. For cells irradiated at the NASA Johnson Space Center facility, the cells were immediately transferred to a bioreactor and attached to a RCCS base rotator in a 370C, 5% CO2 incubator, and the microgravity environment was modeled by rotating the bioreactors at 10 rotations per minute. For cells irradiated at the Brookhaven National Laboratory facility, the cells were shipped to Johnson Space Center the day after irradiation, the cells were received at Johnson Space Center two days after irradiation, and transferred to the bioreactor with a media refresh on the third day. Once transferred to the bioreactor, the bioreactor was attached to a RCCS base rotator in a 370C, 5% CO2 incubator, and the microgravity environment was modeled by rotating the bioreactors at 10 rotations per minute. For the control conditions, cells were grown in a T-75 flask as previously mentioned in Aim 2. Cells were aseptically split and harvested every four days by removing 25 ml of cells and media and replacing it with 25 ml of fresh RPMI 1640 complete media until day 16 for JSC cells and day 20 for BNL cells.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Akata EBV positive cells showed no significant change in cell viability when comparing the microgravity conditions and different radiation doses to control conditions. Akata EBV positive cells also did not show any significant changes in either total cell size (microns) or live cell size (microns) when comparing the microgravity conditions and the different acute radiation doses to the control conditions. The Akata EBV positive cell pellet showed a significant increase on day 12 in the 2.0 Gy bioreactor when compared to the controls. There was also a small increase on day 12 and an even larger increase on day 16 in the 1.0 Gy bioreactor; however, neither of these increases were statistically significant.

Accomplishments towards Aim 5:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 70 ml of media in a T-25 flask, on day 0, for shipment to Brookhaven's National Laboratory (BNL). The cells were irradiated with a proton source with an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy at BNL's facilities on Day 3. On day 6 cells were then split, aliquoted, and fed. This was repeated for an additional four time points. The time points were adjusted for graphing as previously mentioned in Aim 1.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Akata EBV positive cells showed no significant change in cell viability when comparing the different acute proton radiation doses and the control conditions. Akata EBV positive cells from different doses of proton irradiated flasks also did not show a significant change in either total cell size (microns) or live cell size (microns) when compared to the control conditions. The Akata EBV positive cell pellet showed a statistically significant increase in the mean percent change of EBV viral load from day 0 for the 0.5 Gy, 1.0 Gy, and 2.0 Gy proton flasks on day 4, compared to control conditions.

Accomplishments towards Aim 6:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 70 ml of media in a T-25 flask, on day 0, for shipment to Brookhaven's National Laboratory (BNL). The cells were irradiated with a proton source with an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy at BNL's facilities on Day 3. For cells irradiated at the BNL facility, the cells were shipped to Johnson Space Center the day after irradiation, the cells were received at JSC two days after irradiation, and transferred to the bioreactor with a media refresh on the third day (day 6). Once transferred to the bioreactor, the bioreactor was attached to a RCCS base rotator in a 370C, 5% CO2 incubator, and the microgravity environment was modeled by rotating the bioreactors at 10 rotations per minute. Cells were aseptically split and harvested every four days by removing 25 ml of cells and media and replacing it with 25 ml of fresh RPMI 1640 complete media until day 20. The time points were adjusted for graphing as previously mentioned in Aim 1.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Statistical analysis for statistically significant changes in viability, total cell size, and live cell size could not be done since the experiment was done with an N=2 for each condition. An increase in the mean percent change of the EBV viral load from day 0 is observed; however, the experiment was done with only an N=2 so the t-test could not be performed to test for statistical significance.

Accomplishments towards Aim 7:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 70 ml of media in a T-25 flask, on day 0, for shipment to Brookhaven's National Laboratory (BNL). The cells were irradiated with a carbon ion source with an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy at BNL's facilities on Day 3. On day 6 cells were then split, aliquoted, and fed. This was repeated for an additional four time points. The time points were adjusted for graphing as previously mentioned in Aim 1.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Statistical analysis for statistically significant changes in viability, total cell size (microns), and live cell size (microns) could not be done since the experiment was done with an N=2 for each condition. An increase in the mean percent change of the EBV viral load from day 0 is observed on day 4 for the 0.1 Gy, 0.5 Gy, 1.0 Gy, and 2.0 Gy carbon ion irradiated flasks; however, the experiment was done with only an N=2 so the t-test could not be performed to test for statistical significance.

Accomplishments towards Aim 8:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 70 ml of media in a T-25 flask, on day 0, for shipment to Brookhaven's National Laboratory (BNL). The cells were irradiated with a carbon ion source with an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy at BNL's facilities on Day 3. On day 6 cells were transferred to a 50 ml HARV bioreactor and attached to the RCCS base rotator to model the microgravity environment. On day 6 cells were also split, aliquoted, and fed. This was repeated for an additional four time points. The time points were adjusted for graphing as previously mentioned in Aim 1.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Statistical analysis for statistically significant changes in viability, total cell size (microns), and live cell size (microns) could not be done since the experiment was done with an N=2 for each condition. An increase in the mean percent change of the EBV viral load from day 0 is observed on day 4 for the 0.1 Gy, 0.5 Gy, 1.0 Gy, and 2.0 Gy carbon ion irradiated bioreactors; however, the experiment was done with only an N=2 so the t-test could not be performed to test for statistical significance.

Accomplishments towards Aim 9:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 70 ml of media in a T-25 flask, on day 0, for shipment to Brookhaven's National Laboratory (BNL). The cells were irradiated with an iron ion source with an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy at BNL's facilities on Day 3. On day 6 cells were then split, aliquoted, and fed. This was repeated for an additional four time points. The time points were adjusted for graphing as previously mentioned in Aim 1.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Statistical analysis for statistically significant changes in viability, total cell size (microns), and live cell size (microns) could not be done since the experiment was done with an N=2 for each condition. There is a small increase in the mean percent change of the EBV viral load from day 0 is observed on day 4 for the 0.1 Gy, 0.5 Gy, 1.0 Gy, and 2.0 Gy iron ion irradiated flasks; however, the experiment was done with only an N=2 so the t-test could not be performed to test for statistical significance.

Accomplishments towards Aim 10:

Akata EBV Positive cells were seeded at a concentration of 1.0e6 cells/ml in 70 ml of media in a T-25 flask, on day 0, for shipment to Brookhaven's National Laboratory (BNL). The cells were irradiated with an iron ion source with an acute dose of 0.1 Gy, 0.5 Gy, 1.0 Gy, or 2.0 Gy at BNL's facilities on Day 3. On day 6 cells were transferred to a 50 ml HARV bioreactor and attached to the RCCS base rotator to model the microgravity environment. On day 6 cells were also split, aliquoted, and fed. This was repeated for an additional four time points. The time points were adjusted for graphing as previously mentioned in Aim 1.

As previously mentioned in Aim 2, Akata EBV positive cells were evaluated for any changes in cell viability and cell size by an AO/PI stain with the Nexcelom Cellometer. Statistical analysis for statistically significant changes in viability, total cell size (microns), and live cell size (microns) could not be done since the experiment was done with an N=2 for each condition. An increase in the mean percent change of the EBV viral load from day 0 is observed on day 4 for the 0.1 Gy, 0.5 Gy, 1.0 Gy, and 2.0 Gy iron ion irradiated bioreactors; however, the experiment was done with only an N=2 so the t-test could not be performed to test for statistical significance.

PUBLICATIONS

The following article has been submitted to the Journal of Radiation Research:

Satish K. Mehta, Ianik Plante, David C. Bloom, Raymond Stowe, Ashlie Renner, Dhruv Markan, Ye Zhang, Honglu Wu, Brian Crucian, Duane L. Pierson. "Reactivation of latent Epstein-Barr virus; a comparison after gamma rays and proton treatment." Submitted to J of Radiation Research.

Bibliography Type: Description: (Last Updated: 04/15/2025)

Abstracts for Journals and Proceedings

Bloom DC, Renner AN, Dhummakupt A, Stowe RP, Wu H, Mehta SK, Pierson DL. "Roles of Microgravity and Ionizing Radiation on Reactivation of Epstein-Barr Virus in vitro." Presented at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016.
32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016